Lab 4: Requirement Elicitation – I

System Specification: LOANZO (Peer-to-Peer Microfinance and Purpose-Bound Social Lending Platform)

Activities Conducted:

1. Identification and Classification of Stakeholders
2. Elicitation of Functional Requirements
3. Elicitation of Non-Functional Requirements

Document Outcome:

Preliminary Requirements Specification and Prioritization Matrix

1. Identification and Classification of Stakeholders

Stakeholders are categorized based on their operational boundaries, system interactions, and governance responsibilities within the lending ecosystem.

Classification	Stakeholder Group	Operational Role and System Boundary	Core Governance and Technical Expectations
Direct End User	Borrowers (Micro-Enterprises and Consumers)	Submits loan applications, verifies identity, registers collateral, reviews competitive offers, executes legal promissory notes, and conducts installment repayments.	Transparent reducing-balance interest, complete elimination of advance processing fees, flexible tenure alignment, non-coercive collections, and accessible vernacular navigation.
Direct End User	Private Capital Providers (Retail Lenders)	Underwrites micro-loans, submits competitive bids within the marketplace, evaluates counterparty risk profiles, and monitors repayment performance.	Verified counterparty identities, legally binding Promissory Notes compliant with Section 4 of the Negotiable Instruments Act, 1881, collateral backing, and default mitigation.
Field Operations	Certified Field Verification Officers	Independent ground personnel conducting doorstep applicant verification, residential audits, and physical collateral appraisal.	Strict privilege isolation (prohibited from borrowing or lending), structured digital inspection checklists, verified dispatch radiuses, and automated per-visit compensation.
Commercial Partner	Verified Commercial Suppliers and Merchants	Commercial entities (equipment dealers, healthcare facilities, education providers) receiving direct tranche settlements for approved expenditure.	Immediate settlement reconciliation, elimination of chargeback exposure, and automated invoice verification via account validation protocols.
System Administration	Platform Operations and Compliance Management	Platform operators supervising operational integrity, reviewing field officer accreditation, monitoring systemic risk, and handling formal dispute resolution.	Immutable audit trails, automated delinquency alerting, real-time risk assessment telemetry, and strict adherence to statutory lending guidelines.
Regulatory Authority	Financial and Payment Regulators (RBI, NPCI)	Statutory entities enforcing regulatory frameworks concerning digital lending, payment clearing systems, and consumer data protection.	Non-custodial fund architecture (zero customer capital pooling), non-compounding penal rate caps (RBI/2023-24/53), and full adherence to data localization mandates.

2. Elicitation of Functional Requirements

Functional requirements detail the specific business workflows, automated logic, inputs, and operational outputs governing the system.

Identifier	Functional Domain	Requirement Specification and Operational Logic
FR-01	**Identity & Access Management**	The system shall authenticate users via multi-factor credentials, including telephone one-time passwords (OTP), OAuth integrations, and biometric authentication (fingerprint and facial recognition) backed by hardware cryptographic keystores.
FR-02	**Digital Identity Verification**	The system shall interface with DigiLocker to extract verified Aadhaar and Permanent Account Number (PAN) records and execute on-device facial liveness verification via computer vision algorithms.
FR-03	**Cryptographic Data Vault**	The system shall maintain an AES-256 encrypted local repository for sensitive identity documentation, restricting decryption strictly to authenticated sessions verified via biometrics or master credentials.
FR-04	**Role-Based Privilege Isolation**	The system shall enforce strict privilege separation between retail users and field verification officers, explicitly prohibiting verification personnel from engaging in borrowing or capital-lending operations.
FR-05	**Bilateral Loan Origination**	The system shall support bidirectional transaction origination, permitting borrowers to publish structured loan requisitions and lenders to extend direct credit proposals.
FR-06	**Social Marketplace Bidding**	The system shall provide a competitive marketplace allowing verified lenders to submit competing bids on interest rates (Annual Percentage Rate) and repayment tenures.
FR-07	**Amortization Simulator**	The system shall dynamically calculate reducing-balance, simple, and flat interest amortization schedules with visual breakdowns of principal and interest components.

FR-08	**Milestone Tranche Allocation**	The system shall enforce milestone-based capital disbursement, disallowing unmonitored lump-sum disbursements to mitigate capital diversion risks.
FR-09	**Direct Payee Settlement**	The system shall route approved capital disbursements directly to verified commercial vendor accounts via Unified Payments Interface (UPI) upon submission of verified tax invoices.
FR-10	**Penny-Drop Account Verification**	The system shall execute an automated INR 1.00 penny-drop verification routine to ensure beneficiary bank account holder details precisely match the legal vendor identity specified on commercial invoices.
FR-11	**Collateral Asset Registry**	The system shall record, track, and appraise five collateral classes: Precious Bullion (maximum 75% LTV), Real Estate Deeds, Commercial IT Equipment, Motor Vehicles, and Accounts Receivable.
FR-12	**Ground Verification Dispatch**	The system shall dispatch certified field verification officers to capture timestamped, geofenced photographic records of physical assets and business premises.
FR-13	**Dual-Appraisal Consensus**	The system shall enforce independent multi-party inspection protocols for collateral valuations exceeding INR 100,000 to prevent valuation fraud and collusion.
FR-14	**Statutory Contract Synthesis**	The system shall automatically generate legally enforceable Promissory Notes compliant with Section 4 of the Negotiable Instruments Act, 1881, alongside a standardized Key Fact Statement (KFS).
FR-15	**Cryptographic Contract Execution**	The system shall facilitate dual-party digital execution utilizing Aadhaar-based electronic signatures and embed SHA-256 cryptographic audit digests for legal admissibility under Section 65B of the Indian Evidence Act.
FR-16	**Automated UPI Collections**	The system shall generate dynamic UPI QR payloads and deep links (upi://pay) for installment reconciliation and automated ledger updates.
FR-17	**Fair Lending Penal Engine**	The system shall calculate non-compounding daily overdue charges strictly capped at a 2% monthly ceiling in compliance with regulatory fair lending directives.
FR-18	**Distress Restructuring & Settlement**	The system shall record mutually agreed tenure extensions during verified borrower distress and issue digitally signed No Objection Certificates (NOC) upon complete reconciliation.
FR-19	**Conversational AI Advisory**	The system shall incorporate speculative query orchestration across localized language models to deliver vernacular financial explanations.
FR-20	**Automated Event Notification**	The system shall deliver asynchronous lifecycle event notifications, repayment schedules, and dispute notices via secured messaging webhooks and push notification channels.

3. Elicitation of Non-Functional Requirements

Non-functional requirements quantify quality attributes, performance thresholds, architectural constraints, and statutory boundaries.

Identifier	Classification	Specification and Quantitative Benchmark	Architectural Control Mechanism
NFR-01	**Performance**	Local database query execution shall not exceed 15 milliseconds on baseline Android devices (2GB RAM).	Normalized Room SQLite architecture with compiled index DAOs.
NFR-02	**Performance**	Cold application startup to an interactive dashboard state shall execute in under 1.8 seconds with zero visual flicker.	Pre-warmed session initialization evaluating security gates asynchronously.
NFR-03	**Performance**	Conversational advisory first-token response latency shall remain under 1,200 milliseconds over mobile networks.	Speculative parallel query routing with local heuristic fallback.
NFR-04	**Security**	All stored personal data, KYC documents, and local database records shall be encrypted at rest using AES-256-GCM.	Hardware-backed Android KeyStore / StrongBox implementation.
NFR-05	**Security**	All network communications shall be transmitted over TLS 1.3 with mandatory certificate pinning.	OkHttp network interceptor enforcing SHA-256 public key hashes.
NFR-06	**Privacy**	The application shall refrain from accessing contact registries or device media files, adhering strictly to the DPDP Act, 2023.	Minimal permission architecture with isolated application sandboxing.
NFR-07	**Reliability**	Core informational and calculation features shall remain fully operational in zero-connectivity environments.	Local persistence layer with transactional background delta synchronization.
NFR-08	**Availability**	Cloud synchronization interfaces and webhook endpoints shall maintain 99.9% operational availability.	Geographically distributed serverless infrastructure.
NFR-09	**Usability**	The user interface shall support runtime translation across 21 regional languages with transition latencies below 150 milliseconds.	In-memory LRU translation caching.
NFR-10	**Usability**	Critical financial actions shall incorporate deliberate confirmation controls to eliminate unintended execution.	Bilateral draggable confirmation interfaces with haptic feedback.
NFR-11	**Compliance**	The platform shall maintain zero custodial deposit balances, executing 100% of transactions through direct account settlements.	Peer-to-peer settlement architecture compliant with central banking regulations.
NFR-12	**Compliance**	All synthesized digital contracts shall be structured to support summary recovery proceedings under Order XXXVII CPC.	Statutory promissory note formatting and cryptographic timestamping.

Outcome: Preliminary Requirements Specification

Consolidated and prioritized baseline requirements cataloged using the formal MoSCoW prioritization model.

Identifier	Requirement Summary	Type	Target Stakeholder	Priority (MoSCoW)
PRL-01	Multi-Factor & Biometric User Authentication	FR	Borrower, Capital Provider	**Must Have**
PRL-02	DigiLocker KYC & On-Device Facial Liveness Detection	FR	Borrower, Operations	**Must Have**
PRL-03	AES-256 Hardware Encrypted Local Document Vault	NFR	All Users	**Must Have**
PRL-04	Social P2P Reverse-Auction Bidding Marketplace	FR	Borrower, Capital Provider	**Must Have**
PRL-05	Purpose-Bound Direct Merchant Disbursal via UPI	FR	Borrower, Commercial Supplier	**Must Have**
PRL-06	INR 1.00 Penny-Drop Account Name Verification	FR	Commercial Supplier, Capital Provider	**Must Have**
PRL-07	Section 4 NI Act Digital Promissory Note Synthesis	FR	Borrower, Capital Provider	**Must Have**
PRL-08	Aadhaar OTP e-Signing with SHA-256 Tamper Audit Trail	FR	Borrower, Capital Provider	**Must Have**
PRL-09	Dynamic UPI QR & Deep-Link Repayment Settlements	FR	Borrower, Capital Provider	**Must Have**
PRL-10	Non-Compounding Statutory Penal Engine (2% Cap)	FR/NFR	Borrower, Regulatory Authority	**Must Have**
PRL-11	Doorstep Field Officer Dispatch & Physical Verification Desk	FR	Field Operations, Operations	Should Have
PRL-12	Multi-Asset Collateral Escrow Tracking (5 Categories)	FR	Capital Provider, Borrower	Should Have
PRL-13	Offline-First Room SQLite Architecture with WorkManager	NFR	All Users	Should Have
PRL-14	Multi-Model Speculative AI Assistant (Gemini / SambaNova)	FR	Borrower	Should Have
PRL-15	Automated Telegram Alert Desk (@Loanzo_bot) & FCM Push	FR	All Users, Operations	Should Have
PRL-16	Runtime Language Switcher for 21+ Regional Languages	NFR	Borrower	Should Have
PRL-17	Dual-Agent Consensus for Assets Exceeding INR 100,000	FR	Field Operations, Capital Provider	Could Have
PRL-18	Prepayment & Foreclosure Savings Simulation Tool	FR	Borrower	Could Have
PRL-19	Alternative Digital Credit Dossier Generation	FR	Borrower, Commercial Bank	Could Have
PRL-20	Automated Blockchain Smart Contract Mirroring	FR	External Auditors	Deferred (Phase 2)